

(Paper Rough Draft) Tutorial: Comparing Permutativity G -Pseudoalgebra and $\mathcal{F}_G$ -Spaces

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Contents

1	Abstract (ND)	2
2	Introduction (ND)	2
3	Categorical Preliminaries: Category of Small Categories	3
3.1	2-Categories	3
3.2	Chaotic-Forgetful Adjunction	4
3.3	Pseudofunctor, Pseudotransform, and Modification	4
4	Categorical Preliminaries: Operads	7
4.1	Symmetric Monoidal Categories	7
4.2	Operads, Operadic Algebras	8
5	Equivariance Preliminaries (ND)	10
5.1	G -Categories (ND)	10
6	Essential Categorical Tools (ND)	11
6.1	G -Operad $\mathcal{Q}_G$ in $\mathbf{Cat}_G$	11
6.2	The 2-Category $\mathcal{Q}_G\text{-}\mathbf{PsAlg}$	11
6.3	Category $\mathcal{F}_G$ (ND)	15
6.4	2-Category $\mathcal{F}_G\text{-}\mathbf{PsAlg}$	15
7	Construction of $\mathbb{R}_G$ (ND)	17
7.1	Construction of Objects	17
7.2	Extension to 2-functor	17

Here I tried to contain all possible details / definitions. I'll see which ones are supposed to be cut eventually. For some of the stuffs that can be omitted, check the SymMonCat paper for reference.

1 Abstract (ND)

2 Introduction (ND)

Fix G as a finite group. Let P_G be the *associativity G -operad* in $\mathbf{Set}_G$, defined by $P_G(n) := \mathbf{Set}(G, \Sigma_n)$, with structure maps defined by post composition

$$\gamma : \mathbf{Set}(G, \Sigma_k) \times \prod_{r=1}^k \mathbf{Set}(G, \Sigma_{j_r}) \cong \mathbf{Set} \left(G, \Sigma_k \times \prod_{r=1}^k \Sigma_{j_r} \right) \xrightarrow{\gamma \circ -} \mathbf{Set}(G, \Sigma_j),$$

, where $j = \sum_{r=1}^k j_r$, and $\gamma : \Sigma_k \times \prod_{r=1}^k \Sigma_{j_r} \rightarrow \Sigma_j$ is the structure map for associativity operad $\mathcal{M}$ in $\mathbf{Set}$. This is naturally a free $(G \times \Sigma)$ -operad, as the right $(G \times \Sigma_n)$ -action at each $P_G(n)$ is free.

However, it is mentioned in [1] that such G -operad is not finitely generated, as a rough explanation, for every $n \in \mathbb{N}$, there exists $m > n$ and a function $f : G \rightarrow \Sigma_m$, such that it cannot be generated using $\{P_G(k)\}_{k \leq n}$ with the operadic structure map and the $(G \times \Sigma)$ -actions.

As a substitution, [1] defines a finitely-generated suboperad $Q_{G, \mathfrak{D}} \hookrightarrow P_G$ (for some parameter $\mathfrak{D}$ that's associated with choice of transitive G -sets), such that after chaotic generalization the two are equivalent in $\mathbf{Cat}_G$, and having equivalent categories of operadic algebras.

We'll mimic the statement, define an intermediate operad $Q_{G, \mathfrak{D}} \hookrightarrow Q_G \hookrightarrow P_G$, promote it to $\mathcal{Q}_G$ in $\mathbf{Cat}_G$, and *attempt to* modify the proof in [9] to show the equivalence of 2-categories between $\mathcal{Q}_G\text{-PsAlg}$ and $\mathcal{F}_{G, \mathbb{R}}\text{-PsAlg}$.

Future task (assume this works): Prove the equivalence of operads/algebras/pseudoalgebras with the original permutativity G -operad.

Current Goal: Add in extra axioms, so that there exist two 2-functors

$$\mathbb{R}_G : \mathcal{Q}_G\text{-PsAlg} \rightleftarrows \mathcal{F}_{G, \mathbb{R}, -}\text{-PsAlg} : \mathbb{Q}_G$$

inducing an isomorphism of (strict) 2-categories. Here, $-$ is some (yet to be determined) axiom.

3 Categorical Preliminaries: Category of Small Categories

3.1 2-Categories

Definition 3.1. A category $\mathcal{C}$ is a *2-category*, if for each object $X, Y \in \mathcal{C}$ (called *0-cells*), the homset $\mathcal{C}(X, Y)$ is itself a category (called the *hom-category*). Also, for every 0-cells $X, Y, Z \in \mathcal{C}$, the *composition* $\circ : \mathcal{C}(Y, Z) \times \mathcal{C}(X, Y) \rightarrow \mathcal{C}(X, Z)$ is a functor.

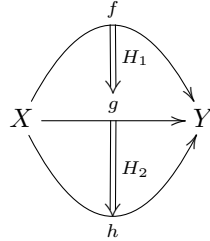
Every morphism $f \in \mathcal{C}(X, Y)$ is called a *1-cell*, while given any 1-cells $f, g \in \mathcal{C}(X, Y)$, a morphism $\alpha : f \Rightarrow g$ in the hom-category $\mathcal{C}(X, Y)$ is called a *2-cell*.

Here, using the unitality and associativity of composition in a category, $\circ$ must be strictly unital and associative.

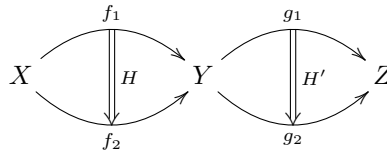
Remark 3.2. There is a more general notion of a *bicategory*, which relaxes the composition functors to be associative and unital up to coherent natural isomorphisms, and in some literature this is called 2-categories. Here we'll take the strict associativity and unitality as axioms for 2-categories.

An important notion is the compositions allowed for the 2-cells. Composition of 2-cells within the same hom-category $\mathcal{C}(X, Y)$ makes sense, as it's a defining feature of a category; on the other hand, since composition $\circ : \mathcal{C}(Y, Z) \times \mathcal{C}(X, Y) \rightarrow \mathcal{C}(X, Z)$ is a functor, any 2-cells $H : f_1 \Rightarrow f_2$ in $\mathcal{C}(X, Y)$ and $H' : g_1 \Rightarrow g_2$ in $\mathcal{C}(Y, Z)$ induces a 2-cell $\circ(H', H) : (g_1 \circ f_1) \Rightarrow (g_2 \circ f_2)$ by inspecting the definition of functors from product category.

Definition 3.3. Given $\mathcal{C}$ a 2-category, the *vertical composition* of 2-cells is the composition within each hom-category $\mathcal{C}(X, Y)$: Given 2-cells $H_1 : f \Rightarrow g$ and $H_2 : g \Rightarrow h$ in the hom-category $\mathcal{C}(X, Y)$, the vertical composition is $H_2 H_1 : f \Rightarrow h$:



The *horizontal composition* of 2-cells is the one induced by the composition functor $\circ : \mathcal{C}(Y, Z) \times \mathcal{C}(X, Y) \rightarrow \mathcal{C}(X, Z)$: Given 2-cells $H : f_1 \Rightarrow f_2$ in $\mathcal{C}(X, Y)$ and $H' : g_1 \Rightarrow g_2$ in $\mathcal{C}(Y, Z)$, define their horizontal composition as $H' * H := \circ(H', H) : (g_1 \circ f_1) \Rightarrow (g_2 \circ f_2)$:



By inspecting the properties of composition functor, one realizes that vertical and horizontal composition commutes: Suppose $H_1 : f \Rightarrow g$, $H_2 : g \Rightarrow h$ are 2-cells in $\mathcal{C}(X, Y)$, while $H'_1 : f' \Rightarrow g'$, $H'_2 : g' \Rightarrow h'$ are 2-cells in $\mathcal{C}(Y, Z)$, then

$$(H'_2 H'_1) * (H_2 H_1) = (H'_2 * H_2)(H'_1 * H_1) : (f' \circ f) \Rightarrow (h' \circ h) \quad (3.4)$$

as 2-cells.

Example 3.5. Let $\mathbf{Cat}$ denote the *category of small categories*, where the objects are small categories, the homsets $\mathbf{Cat}(\mathcal{A}, \mathcal{B})$ are functors between small categories $\mathcal{A}$ and $\mathcal{B}$. Notice the homset also equips with a category structure, by specifying the morphisms in $\mathbf{Cat}(\mathcal{A}, \mathcal{B})$ to be natural transformations between functors.

This specifies $\mathbf{Cat}$ to be a 2-category. Also, a precise description of the horizontal composition of 2-cells is as follow: Given functors $F_1, F_2 \in \mathbf{Cat}(\mathcal{A}, \mathcal{B})$ and $G_1, G_2 \in \mathbf{Cat}(\mathcal{B}, \mathcal{C})$, natural transformations $\mu : F_1 \Rightarrow F_2$ in $\mathbf{Cat}(\mathcal{A}, \mathcal{B})$ and $\nu : G_1 \Rightarrow G_2$ in $\mathbf{Cat}(\mathcal{B}, \mathcal{C})$, then $\nu * \mu := \nu \circ F\mu$ defines a horizontal composition.

3.2 Chaotic-Forgetful Adjunction

Given any small category $\mathcal{A} \in \mathbf{Cat}$, its object $\text{Ob}(\mathcal{A})$ forms a set; moreover, any functor $F : \mathcal{A} \rightarrow \mathcal{B}$ also has an underlying map of objects $F : \text{Ob}(\mathcal{A}) \rightarrow \text{Ob}(\mathcal{B})$. Conversely, given a set, it has certain category structure:

Definition 3.6. Given a set X , define $\mathcal{E}X$ to be the *chaotic category* with underlying object set $\text{Ob}(\mathcal{E}X) := X$, by encoding a unique isomorphism $x \xrightarrow{\sim} y$ between any element $x, y \in X$. The composition

$$(x \xrightarrow{\sim} y \xrightarrow{\sim} z) = (x \xrightarrow{\sim} z) \quad (3.7)$$

using the uniqueness.

Given a set map $f : X \rightarrow Y$, define the functor $\mathcal{E}f : \mathcal{E}X \rightarrow \mathcal{E}Y$ so that the underlying object map is f , and for each morphism $x \xrightarrow{\sim} y$ in $\mathcal{E}X$, $\mathcal{E}f(x \rightarrow y) := f(x) \rightarrow f(y)$, the unique morphism between $f(x), f(y)$ in $\mathcal{E}Y$.

Inspecting the definition, we see $\mathcal{E} : \mathbf{Set} \rightarrow \mathbf{Cat}$ is a well-defined functor, called the *chaotic functor*.

There is a well-known adjunction $(\text{Ob} \dashv \mathcal{E})$, due to the fact that after fixing the objects, the morphisms can only be mapped to the unique morphisms between any two objects when the codomain is a chaotic category.

3.3 Pseudofunctor, Pseudotransform, and Modification

In some cases, the notion of functor is too restrictive: strictly preserving f

Definition 3.8 (Pseudofunctor). Given two 2-categories $\mathcal{A}, \mathcal{B}$, a *pseudofunctor* $F : \mathcal{A} \rightarrow \mathcal{B}$ equips with an 0-cell map, functors between hom-categories $F_{X,Y} : \mathcal{A}(X, Y) \rightarrow \mathcal{B}(F(X), F(Y))$, an invertible 2-cell $\phi_{X,Y,Z}$ between compositions

$$\begin{array}{ccc} \mathcal{A}(Y, Z) \times \mathcal{A}(X, Y) & \xrightarrow{F_{Y,Z} \times F_{X,Y}} & \mathcal{B}(F(Y), F(Z)) \times \mathcal{B}(F(X), F(Y)) \\ \downarrow \circ & \nearrow \phi_{X,Y,Z} & \downarrow \circ \\ \mathcal{A}(X, Z) & \xrightarrow{F_{X,Z}} & \mathcal{B}(F(X), F(Z)) \end{array}$$

and another invertible 2-cell ψ_X between identities

$$\begin{array}{ccc}
 & \text{id}_{F(X)} & \\
 & \downarrow \psi_X & \\
 F(X) & \xrightarrow{\quad} & F(X) \\
 & \uparrow F_{X,X}(\text{id}_X) &
 \end{array}$$

such that the following axioms hold:

(i) **(Associativity)** The following two pasting diagrams are equal:

$$\begin{array}{ccc}
 \mathcal{A}(Z, W) \times \mathcal{A}(Y, Z) \times \mathcal{A}(X, Y) & \xrightarrow{\Pi F_{-, -}} & \mathcal{B}(F(Z), F(W)) \times \mathcal{B}(F(Y), F(Z)) \times \mathcal{B}(F(X), F(Y)) \\
 \downarrow \circ \times \text{id} & \swarrow \phi_{Y, Z, W} \times \text{id} & \downarrow \circ \times \text{id} \\
 \mathcal{A}(Y, W) \times \mathcal{A}(X, Y) & \xrightarrow{F_{Y, W} \times F_{X, Y}} & \mathcal{B}(F(Y), F(W)) \times \mathcal{B}(F(X), F(Y)) \\
 \downarrow \circ & \swarrow \phi_{X, Y, W} & \downarrow \circ \\
 \mathcal{A}(X, W) & \xrightarrow{F_{X, W}} & \mathcal{B}(F(X), F(W))
 \end{array}$$

equals

$$\begin{array}{ccc}
 \mathcal{A}(Z, W) \times \mathcal{A}(Y, Z) \times \mathcal{A}(X, Y) & \xrightarrow{\Pi F_{-, -}} & \mathcal{B}(F(Z), F(W)) \times \mathcal{B}(F(Y), F(Z)) \times \mathcal{B}(F(X), F(Y)) \\
 \downarrow \text{id} \times \circ & \swarrow \text{id} \times \phi_{X, Y, Z} & \downarrow \text{id} \times \circ \\
 \mathcal{A}(Z, W) \times \mathcal{A}(X, Z) & \xrightarrow{F_{Z, W} \times F_{X, Z}} & \mathcal{B}(F(Z), F(W)) \times \mathcal{B}(F(X), F(Z)) \\
 \downarrow \circ & \swarrow \phi_{X, Z, W} & \downarrow \circ \\
 \mathcal{A}(X, W) & \xrightarrow{F_{X, W}} & \mathcal{B}(F(X), F(W))
 \end{array}$$

(ii) **(Unitality)** The following two unit diagrams commute:

$$\begin{array}{ccc}
 \mathcal{A}(X, Y) \times \mathcal{A}(X, X) & \xrightarrow{F_{X, Y} \times F_{X, X}} & \mathcal{B}(F(X), F(Y)) \times \mathcal{B}(F(X), F(X)) \\
 \uparrow \text{id} \times \text{id}_X & \nearrow \text{id} * \psi_X & \downarrow \circ \\
 \mathcal{A}(X, Y) \times * & & \\
 \downarrow F_{X, Y} \times \text{id}_{F(X)} & \nwarrow & \downarrow \\
 \mathcal{B}(F(X), F(Y)) \times \mathcal{B}(F(X), F(X)) & \xrightarrow{\quad \circ \quad} & \mathcal{B}(F(X), F(Y))
 \end{array}$$

$$\begin{array}{ccc}
 \mathcal{A}(Y, Y) \times \mathcal{A}(X, Y) & \xrightarrow{F_{Y, Y} \times F_{X, Y}} & \mathcal{B}(F(Y), F(Y)) \times \mathcal{B}(F(X), F(Y)) \\
 \uparrow \text{id}_Y \times \text{id} & \nearrow \psi_Y * \text{id} & \downarrow \circ \\
 * \times \mathcal{A}(X, Y) & & \\
 \downarrow \text{id}_{F(Y)} \times F_{X, Y} & \nwarrow & \downarrow \\
 \mathcal{B}(F(Y), F(Y)) \times \mathcal{B}(F(X), F(Y)) & \xrightarrow{\quad \circ \quad} & \mathcal{B}(F(X), F(Y))
 \end{array}$$

where $\text{id}_X : * \rightarrow \mathcal{A}(X, X)$ is the functor indicating identity of X .

If ψ_X is identity 2-cell for all 0-cell X , then the pseudofunctor F is called *normal*.

Definition 3.9 (Pseudotransform). Fix 2-categories $\mathcal{A}, \mathcal{B}$, let $(F, \phi, \psi), (G, \varphi, \vartheta) : \mathcal{A} \rightarrow \mathcal{B}$ be two pseudofunctors. A *pseudotransform* $\mu : (F, \phi, \psi) \rightarrow (G, \varphi, \vartheta)$ collects 1-cells $\mu_X : F(X) \rightarrow G(X)$ for each 0-cell $X \in \mathcal{A}$, and for each 1-cell $f : X \rightarrow Y$ in $\mathcal{A}$, associates a 2-cell μf such that

$$\begin{array}{ccc} F(X) & \xrightarrow{\mu_X} & G(X) \\ Ff \downarrow & \mu f \swarrow & \downarrow Gf \\ F(Y) & \xrightarrow{\mu_Y} & G(Y) \end{array}$$

commutes.

Furthermore, they satisfy the following axioms:

- (i) **(Associativity)** Let $f : X \rightarrow Y, g : Y \rightarrow Z$ be 1-cells in $\mathcal{A}$, the following two pasting diagrams are equal:

$$\begin{array}{ccc} F(X) & \xrightarrow{\mu_X} & G(X) \\ \downarrow Ff & \searrow \mu f & \downarrow Gf \\ F(Y) & \xrightarrow{\mu_Y} & G(Y) \\ \downarrow F(g \circ f) & \swarrow \phi_{g,f} & \downarrow G(g \circ f) \\ F(Z) & \xrightarrow{\mu_Z} & G(Z) \end{array} \quad = \quad \begin{array}{ccc} F(X) & \xrightarrow{\mu_X} & G(X) \\ \downarrow F(g \circ f) & \searrow \mu(g \circ f) & \downarrow G(g \circ f) \\ F(Z) & \xrightarrow{\mu_Z} & G(Z) \\ \downarrow Ff & \swarrow \phi_{g,f} & \downarrow Gf \\ F(Y) & \xrightarrow{\mu_Y} & G(Y) \end{array}$$

- (ii) **(Unitality)** The following pasting diagrams are equal:

$$\begin{array}{ccc} F(X) & \xrightarrow{\mu_X} & G(X) \\ \downarrow F(\text{id}_X) & \searrow \psi_X & \downarrow \text{id}_{F(X)} \\ F(X) & \xrightarrow{\mu_X} & G(X) \end{array} \quad = \quad \begin{array}{ccc} F(X) & \xrightarrow{\mu_X} & G(X) \\ \downarrow F(\text{id}_X) & \searrow \mu(\text{id}_X) & \downarrow \text{id}_{G(X)} \\ F(X) & \xrightarrow{\mu_X} & G(X) \end{array}$$

Definition 3.10 (Modification). Continue with the same notation as above. Let $\mu, \nu : (F, \phi, \psi) \rightarrow (G, \varphi, \vartheta)$ be two pseudotransforms, a *modification* $\lambda : \mu \Rightarrow \nu$ is a collection of 2-cells $\lambda_X : \mu_X \Rightarrow \nu_X$ for each 0-cell $X \in \mathcal{A}$, such that the following two pasting diagrams are equal, for each 1-cell $f : X \rightarrow Y$ in $\mathcal{A}$:

$$\begin{array}{ccc} F(X) & \xrightarrow{\mu_X} & G(X) \\ \downarrow Ff & \searrow \lambda_X & \downarrow Gf \\ F(Y) & \xrightarrow{\mu_Y} & G(Y) \end{array} \quad = \quad \begin{array}{ccc} F(X) & \xrightarrow{\mu_X} & G(X) \\ \downarrow Ff & \searrow \mu f & \downarrow Gf \\ F(Y) & \xrightarrow{\mu_Y} & G(Y) \end{array}$$

4 Categorical Preliminaries: Operads

4.1 Symmetric Monoidal Categories

Definition 4.1. A category $\mathcal{V}$ is a *monoidal category*, if there is a bifunctor $\otimes : \mathcal{V} \times \mathcal{V} \rightarrow \mathcal{V}$ (called *monoidal product*), and an object $\kappa \in \mathcal{V}$ (called *unit/monoidal unit*), together with the following natural isomorphisms:

- The *associator* $a : (- \otimes -) \otimes - \xrightarrow{\sim} - \otimes (- \otimes -)$ of functors $\mathcal{V}^3 \xrightarrow{\sim} \mathcal{V}$.
- The *left unit* $\lambda : \kappa \otimes - \xrightarrow{\sim} \text{Id}_{\mathcal{V}}$ and the *right unit* $\rho : - \otimes \kappa \xrightarrow{\sim} \text{Id}_{\mathcal{V}}$ of functors $\mathcal{V} \rightarrow \mathcal{V}$.
- The *Symmetric Braiding* $B : -_1 \otimes -_2 \xrightarrow{\sim} -_2 \otimes -_1$, with extra property $B_{Y,X} : Y \otimes X \rightarrow X \otimes Y$ equals to $B_{X,Y}^{-1} : X \otimes Y \rightarrow Y \otimes X$.

They also satisfy the following axioms:

- *Triangle Axiom:*

$$\begin{array}{ccc} (X \otimes \kappa) \otimes Y & \xrightarrow{a_{X,\kappa,Y}} & X \otimes (\kappa \otimes Y) \\ \rho_X \otimes \text{id}_Y \searrow & & \swarrow \text{id}_X \otimes \lambda_Y \\ & X \otimes Y & \end{array}$$

- *Pentagon Axiom:*

$$\begin{array}{ccc} ((X \otimes Y) \otimes Z) \otimes W & \xrightarrow{a_{X \otimes Y, Z, W}} & (X \otimes Y) \otimes (Z \otimes W) \\ \downarrow a_{X,Y,Z} \otimes \text{id}_W & & \downarrow a_{X,Y,Z \otimes W} \\ (X \otimes (Y \otimes Z)) \otimes W & & X \otimes (Y \otimes (Z \otimes W)) \\ \searrow a_{X,Y \otimes Z, W} & & \swarrow \text{id}_X \otimes a_{Y,Z,W} \\ & X \otimes ((Y \otimes Z) \otimes W) & \end{array}$$

- *Hexagon Axiom:*

$$\begin{array}{ccccc} (X \otimes Y) \otimes Z & \xrightarrow{a_{X,Y,Z}} & X \otimes (Y \otimes Z) & \xrightarrow{B_{X,Y \otimes Z}} & (Y \otimes Z) \otimes X \\ \downarrow B_{X,Y} \otimes \text{id}_Z & & & & \downarrow a_{Y,Z,X} \\ (Y \otimes X) \otimes Z & \xrightarrow{a_{Y,X,Z}} & Y \otimes (X \otimes Z) & \xrightarrow{\text{id}_Y \otimes B_{X,Z}} & Y \otimes (Z \otimes X) \\ & & & & \\ X \otimes (Y \otimes Z) & \xleftarrow{a_{X,Y,Z}} & (X \otimes Y) \otimes Z & \xrightarrow{B_{X \otimes Y, Z}} & Z \otimes (X \otimes Y) \\ \downarrow \text{id}_X \otimes B_{Y,Z} & & & & \uparrow a_{Z,X,Y} \\ X \otimes (Z \otimes Y) & \xleftarrow{a_{X,Z,Y}} & (X \otimes Z) \otimes Y & \xrightarrow{B_{X,Z \otimes Y}} & (Z \otimes X) \otimes Y \end{array}$$

Based on *MacLane's Coherence Theorem*, the above two axioms guarantees all finite sequences of a, λ, ρ and their inverses agree, namely there is a unique isomorphism $P_1 \rightarrow P_2$, where P_i are (possibly) different orders of performing product on $X_1, \dots, X_n \in \mathcal{V}$. For reference, cf. *Category Theory for Working Mathematicians*.

4.2 Operads, Operadic Algebras

Definition 4.2. Given symmetric monoidal category $\mathcal{V}$, a Σ *operad* $\mathcal{C}$ in $\mathcal{V}$ is a collection of objects $\{\mathcal{C}(j) \in \mathcal{V}\}_{j \in \mathbb{N}}$, each $\mathcal{C}(j)$ with a right Σ_j -action (a group homomorphism $\Sigma_j^{\text{op}} \rightarrow \text{Aut}_{\mathcal{V}}(\mathcal{C}(j))$), a *unit map* $\eta : 1 \rightarrow \mathcal{C}(1)$, together with composition maps

$$\gamma : \mathcal{C}(k) \otimes \mathcal{C}(j_1) \otimes \dots \otimes \mathcal{C}(j_k) \rightarrow \mathcal{C}(j_1 + \dots + j_k)$$

that satisfies the following axioms:

- *Unit:*

$$\begin{array}{ccc} \mathcal{C}(j) \otimes \kappa^{\otimes j} & \xrightarrow{\sim} & \mathcal{C}(j) \\ \searrow \text{id}_{\mathcal{C}(j)} \otimes \eta^{\otimes j} & & \nearrow \gamma \\ & \mathcal{C}(j) \otimes \mathcal{C}(1)^{\otimes j} & \end{array} \quad \begin{array}{ccc} \kappa \otimes \mathcal{C}(j) & \xrightarrow{\sim} & \mathcal{C}(j) \\ \searrow \eta \otimes \text{id}_{\mathcal{C}(j)} & & \nearrow \gamma \\ & \mathcal{C}(1) \otimes \mathcal{C}(j) & \end{array}$$

- *Associativity:* Fix natural numbers $k, j = \sum_{s=1}^k j_s$, define index $g_s := j_1 + \dots + j_s$ for $1 \leq s \leq k$. For each $i_1, \dots, i_{g_k} = i_j$, with sum $i = \sum_{r=1}^j i_r$, and $h_s := i_{g_{(s-1)+1}} + \dots + i_{g_s}$ (so $i = \sum_{s=1}^k h_s$ also), the following diagram commutes:

$$\begin{array}{ccc} \left[\mathcal{C}(k) \otimes \left(\bigotimes_{s=1}^k \mathcal{C}(j_s) \right) \right] \otimes \left(\bigotimes_{r=1}^j \mathcal{C}(i_r) \right) & \xrightarrow{\gamma \otimes \text{id}} & \mathcal{C}(j) \otimes \left(\bigotimes_{r=1}^j \mathcal{C}(i_r) \right) \\ \downarrow \text{shuffle} & & \downarrow \gamma \\ \mathcal{C}(k) \otimes \left[\bigotimes_{s=1}^k \left(\mathcal{C}(j_s) \otimes \left(\bigotimes_{q=1}^{j_s} \mathcal{C}(i_{g_{(s-1)+q}}) \right) \right) \right] & \xrightarrow{\text{id}_{\mathcal{C}(k)} \otimes \left(\bigotimes_s \gamma \right)} & \mathcal{C}(k) \otimes \left(\bigotimes_{s=1}^k \mathcal{C}(h_s) \right) \\ & & \uparrow \gamma \\ & & \mathcal{C}(i) \end{array}$$

- *Equivariance:*

- Given $\sigma \in \Sigma_k$, $j_1 + \dots + j_k = \Sigma_j$, let $\sigma(j_1, \dots, j_k) \in \Sigma_j$ denotes “permutation of k -blocks of letters”, by $(j_1, \dots, j_k) \mapsto (j_{\sigma^{-1}(1)}, \dots, j_{\sigma^{-1}(k)})$.

Example 4.3. Given $j = 5 = 2 + 3$, the permutation $(1 \ 2) \in \Sigma_2$, $\sigma(2, 3)$ is the permutation $\langle 1, 2; 3, 4, 5 \rangle \mapsto \langle 3, 4, 5; 1, 2 \rangle$ (namely changing the blocks’ order, without changing the order in each block).

Then, the following diagram commutes:

$$\begin{array}{ccc} \mathcal{C}(k) \otimes \left(\bigotimes_{r=1}^k \mathcal{C}(j_r) \right) & \xrightarrow{\sigma \otimes \text{shuffle}_{\sigma^{-1}}} & \mathcal{C}(k) \otimes \left(\bigotimes_{r=1}^k \mathcal{C}(j_{\sigma(r)}) \right) \\ \downarrow \gamma & & \downarrow \gamma \\ \mathcal{C}(j) & \xrightarrow{\sigma(j_{\sigma(1)}, \dots, j_{\sigma(k)})} & \mathcal{C}(j) \end{array}$$

- Given $\tau_r \in \Sigma_{j_r}$, denote $\tau_1 \oplus \dots \oplus \tau_k \in \Sigma_j$ by letting each τ_r permuting the j_r th block in j (for instance, let τ_1 acts on $1, \dots, j_1$; let τ_2 acts on $j_1 + 1, \dots, j_1 + j_2$, etc.). Then, the following diagram

commutes:

$$\begin{array}{ccc}
\mathcal{C}(k) \otimes \left(\bigotimes_{r=1}^k \mathcal{C}(j_r) \right) & \xrightarrow{\text{id}_{\mathcal{C}(k)} \otimes (\bigotimes_r \tau_r)} & \mathcal{C}(k) \otimes \left(\bigotimes_{r=1}^k \mathcal{C}(j_r) \right) \\
\downarrow \gamma & & \downarrow \gamma \\
\mathcal{C}(j) & \xrightarrow{\tau_1 \oplus \dots \oplus \tau_k} & \mathcal{C}(j)
\end{array}$$

If $\mathcal{C}(0) = \kappa$ (with $\gamma : \mathcal{C}(0) \rightarrow \mathcal{C}$ being $\text{id}_{\mathcal{C}(0)}$), the operad is *reduced*.

Definition 4.4. Let $\mathcal{C}, \mathcal{C}'$ be two operads on symmetric monoidal category $\mathcal{V}$, a *morphism of operads* $\psi : \mathcal{C} \rightarrow \mathcal{C}'$, is a sequence of Σ_j -equivariant morphisms $\psi_j : \mathcal{C}(j) \rightarrow \mathcal{C}'(j)$ in $\mathcal{V}$ (i.e., $\sigma^{-1} \circ \psi_j \circ \sigma = \psi_j$, for all $\sigma \in \Sigma_j$), such that it preserves the operad's structure morphism:

$$\begin{array}{ccc}
\mathcal{C}(k) \otimes \mathcal{C}(j_1) \otimes \dots \otimes \mathcal{C}(j_k) & \xrightarrow{\gamma} & \mathcal{C}(j_1 + \dots + j_k) \\
\downarrow \psi_k \otimes \psi_{j_1} \otimes \dots \otimes \psi_{j_k} & & \downarrow \psi_{j_1 + \dots + j_k} \\
\mathcal{C}'(k) \otimes \mathcal{C}'(j_1) \otimes \dots \otimes \mathcal{C}'(j_k) & \xrightarrow{\gamma'} & \mathcal{C}'(j_1 + \dots + j_k)
\end{array}$$

Definition 4.5. Given an operad $\mathcal{C}$ over a symmetric monoidal category $\mathcal{V}$, a $\mathcal{C}$ -*algebra* is an object $A \in \mathcal{V}$, with collections of morphisms $\theta : \mathcal{C}(j) \otimes A^j \rightarrow A$, that satisfies the following axioms:

- *Associativity:* Given $j = \sum_{i=1}^k j_i$, the following diagram commutes:

$$\begin{array}{ccc}
\left[\mathcal{C}(k) \otimes \left(\bigotimes_{i=1}^k \mathcal{C}(j_i) \right) \right] \otimes A^{\otimes j} & \xrightarrow{\gamma \otimes \text{id}_{A^{\otimes j}}} & \mathcal{C}(j) \otimes A^{\otimes j} \\
\downarrow \text{shuffle} & & \searrow \theta \\
\mathcal{C}(k) \otimes \left[\bigotimes_{i=1}^k (\mathcal{C}(j_i) \otimes A^{\otimes j_i}) \right] & \xrightarrow{\text{id}_{\mathcal{C}(k)} \otimes \theta^k} & \mathcal{C}(k) \otimes A^{\otimes k} \\
& & \nearrow \theta \\
& & A
\end{array}$$

(Note: This part uses the isomorphism $A^{\otimes j} \cong \bigotimes_{i=1}^k A^{\otimes j_i}$).

- *Unit:* The following diagram commutes:

$$\begin{array}{ccc}
\kappa \otimes A & \xrightarrow{\sim} & A \\
\searrow \eta \otimes \text{id}_A & & \nearrow \theta \\
& C(1) \otimes A &
\end{array}$$

- *Equivariance:* Given any permutation $\sigma \in \Sigma_j$, the following diagram commutes:

$$\begin{array}{ccc}
\mathcal{C}(j) \otimes A^{\otimes j} & \xrightarrow{\sigma \otimes \sigma^{-1}} & \mathcal{C}(j) \otimes A^{\otimes j} \\
\searrow \theta & & \swarrow \theta \\
& A &
\end{array}$$

5 Equivariance Preliminaries

5.1 G -Categories

Definition 5.1. A small category $\mathcal{A}$ is a G -category, if the object and morphism sets $\text{Ob}(\mathcal{A}), \text{Mor}(\mathcal{A})$ are both (left) G -sets, such that the following four structure maps (S, T, I, C) (stands for *source, target, identity, and composition*) are all G -equivariant:

- $S, T : \text{Mor}(\mathcal{A}) \rightarrow \text{Ob}(\mathcal{A})$ by $S(X \xrightarrow{f} Y) := X$, and $T(X \xrightarrow{f} Y) := Y$.
- $I : \text{Ob}(\mathcal{A}) \rightarrow \text{Mor}(\mathcal{A})$ by $I(X) := \text{id}_X$.
- $C : \text{Mor}(\mathcal{A}) \times_{S,T} \text{Mor}(\mathcal{A}) \rightarrow \text{Mor}(\mathcal{A})$ by $C(g, f) := g \circ f$.

Here, $\text{Mor}(\mathcal{A}) \times_{S,T} \text{Mor}(\mathcal{A}) := \{(g, f) \in \text{Mor}(\mathcal{A})^2 \mid S(g) = T(f)\}$, precisely the pair of morphisms that're composable, and it endows G -set structure via restricting the diagonal G -action on $\text{Mor}(\mathcal{A})^2$. This is well-defined due to the assumption S, T are G -maps.

Another equivalent characterization can be viewed as a group homomorphism $G \rightarrow \text{Aut}_{\mathbf{Cat}}(\mathcal{A})$, which assigns each $g \in G$ an automorphism functor on $\mathcal{A}$ that respects G 's multiplication structure, and identity gets sent to the identity functor. So, from now on we view each element $g \in G$ as a functor, whenever a category is specified as a G -category.

Definition 5.2. Let $\mathbf{Cat}_G$ denotes the 2-category of small G -categories, with (nonequivariant) functors as 1-cells, natural transformations as 2-cells. Which, the G -fixed subcategory of the hom-category $[\mathbf{Cat}_G(\mathcal{A}, \mathcal{B})]^G$ precisely describes all G -functor, or the functors $F : \mathcal{A} \rightarrow \mathcal{B}$ such that $g \circ F = F \circ g$ for all $g \in G$.

Another important notion is the twisted G -action on products when given a group homomorphism $\alpha : G \rightarrow \Sigma_n$. Given a category $\mathcal{A}$, the n -fold product $\mathcal{A}^{\mathbf{n}}$ has a natural left Σ_n -action via permuting the coordinates; if $\mathcal{A}$ is a G -category, $\mathcal{A}^{\mathbf{n}}$ also endows a natural diagonal G -action. The two coordinates to give a natural left $(G \times \Sigma_n)$ -action on $\mathcal{A}^{\mathbf{n}}$.

Definition 5.3. Let $\mathcal{A}^{\mathbf{n}^\alpha}$ be a G -category with the underlying category being $\mathcal{A}^{\mathbf{n}}$, with the G -action defined as follow:

$$g \cdot_\alpha (X_1, \dots, X_n) := (g \cdot X_{\alpha(g)^{-1}(1)}, \dots, g \cdot X_{\alpha(g)^{-1}(n)})$$

Another characterization can be understood as follow: Given the graph subgroup $\Gamma_\alpha := \{(g, \alpha(g)) \in G \times \Sigma_n\}_{g \in G}$, the projection $\pi_G : G \times \Sigma_n \rightarrow G$ restricts to an isomorphism $\Gamma_\alpha \xrightarrow{\sim} G$. The α -twisted action is induced by the pullback along this isomorphism.

6 Essential Categorical Tools (ND)

Define P_G , its chaotic categorification first Here we introduce some definitions important to this paper.

6.1 G -Operad $\mathcal{Q}_G$ in $\mathbf{Cat}_G$

Definition 6.1. Let Q_G be the suboperad of P_G , generated by all the group homomorphisms $\alpha : G \rightarrow \Sigma_n$ spanning through all $n \in \mathbb{N}$, the P_G structure maps γ_G , and the $(G \times \Sigma)$ -action on the elements. Define the operad $\mathcal{Q}_G := \mathcal{E}Q_G$ as the chaotic generalization in $\mathbf{Cat}_G$.

This operad is designed as a smallest suboperad that contains all group homomorphisms from G to a symmetric group, such that it's still a free $(G \times \Sigma)$ -operad.

Potential (fun) question: Is this suboperad finitely generated, or equals to any $\mathcal{Q}_{G,\mathfrak{D}}$ described in [1]?

Warning 6.2. If naively taken all group homomorphisms and their orbits under $(G \times \Sigma)$ -action, it doesn't form a suboperad in general for nontrivial group G . A counterexample can be constructed even for $G = \mathbb{Z}/2\mathbb{Z}$.

6.2 The 2-Category $\mathcal{Q}_G\text{-PsAlg}$

We mostly follow [9] in defining $\mathcal{Q}_G$ -pseudoalgebras.

Definition 6.3 ($\mathcal{Q}_G$ -pseudoalgebra). A $\mathcal{Q}_G$ -pseudoalgebra is a G -category $\mathcal{A}$ and operad action functors

$$\theta_j : \mathcal{Q}_G(j) \times \mathcal{A}^j \rightarrow \mathcal{A}$$

for each $n \in \mathbb{N}$, together with an invertible natural transformation ϕ_1 in the diagram

$$\begin{array}{ccc} \mathcal{A} & \xrightarrow{\iota} & \mathcal{Q}_G(1) \times \mathcal{A} \\ & \searrow \phi_1 & \downarrow \theta_1 \\ & \text{Id}_{\mathcal{A}} & C \end{array} \quad (6.4)$$

An invertible natural transformation $\phi = \phi(k; j_1, \dots, j_k)$ for each diagram

$$\begin{array}{ccc} \mathcal{Q}_G(k) \times \left(\prod_{r=1}^k \mathcal{Q}_G(j_r) \times \mathcal{A}^{j_r} \right) & \xrightarrow{\text{id} \times \prod_r \theta_{j_r}} & \mathcal{Q}_G(k) \times \mathcal{A}^k \\ \downarrow \text{shuffle} & \nearrow \phi & \downarrow \theta_k \\ \mathcal{Q}_G(k) \times \prod_{r=1}^k \mathcal{Q}_G(j_r) \times \mathcal{A}^{j_r} & \xrightarrow{\gamma \times \text{id}} & \mathcal{Q}_G(j_+) \times \mathcal{A}^{j_+} \end{array} \quad (6.5)$$

such that the following strict equivariance hold for θ_j and ϕ :

(i) **(Equivariance of θ_j)** The following diagram commutes for $(g, \sigma) \in G \times \Sigma_j$:

$$\begin{array}{ccc} \mathcal{Q}_G(j) \times \mathcal{A}^j & \xrightarrow{(g, \sigma) \times (g, \sigma)^{-1}} & \mathcal{Q}_G(j) \times \mathcal{A}^j \\ \theta_j \downarrow & & \downarrow \theta_j \\ \mathcal{A} & \xrightarrow{g^{-1}} & \mathcal{A} \end{array} \quad (6.6)$$

I think if assuming the commutation for G also, the case $j = 1$ implies θ_j collapses all G -action on $\mathcal{A}$ to be trivial

(ii) **(Equivariance of ϕ)** The following equalities hold when given $g \in G$, $\sigma \in \Sigma_k$, and $\tau_r \in \Sigma_{j_r}$:

$$\phi(k; j_1, \dots, j_k) = \phi(k; j_{\sigma(1)}, \dots, j_{\sigma_k}) \circ ((g, \sigma) \times (g, \sigma)^{-1})$$

$$\phi(k; j_1, \dots, j_k) = \phi(k; j_1, \dots, j_k) \circ (\text{id} \times \prod_r ((g, \tau_r) \times (g, \tau_r)^{-1}))$$

They also need to satisfy the following coherence properties:

(iii) **(Operadic Composition)** The following pasting diagrams are equal, given $j_+ = \sum j_r$, $i_r = \sum_s i_{r,s}$, and $i_+ = \sum_{r,s} i_{r,s}$ with $1 \leq r \leq k$ and $1 \leq s \leq j_r$:

$$\begin{array}{ccccc}
\mathcal{Q}_G(k) \times \prod_r (\mathcal{Q}_G(j_r) \times \prod_s (\mathcal{Q}_G(i_{r,s}) \times \mathcal{A}^{i_{r,s}})) & \xrightarrow{\text{id} \times \prod (\text{id} \times \theta_{i_{r,s}})} & \mathcal{Q}_G(k) \times \prod_r (\mathcal{Q}_G(j_r) \times \mathcal{A}^{j_r}) & \xrightarrow{\text{id} \times \prod \theta_{j_r}} & \mathcal{Q}_G(k) \times \mathcal{A}^k \\
\downarrow \gamma \times \text{id} & & \downarrow \gamma & \searrow \phi & \downarrow \theta_k \\
\mathcal{Q}_G(j_+) \times \prod_{r,s} (\mathcal{Q}_G(i_{r,s}) \times \mathcal{A}^{i_{r,s}}) & \xrightarrow{\text{id} \times \prod \theta_{i_{r,s}}} & \mathcal{Q}_G(j_+) \times \mathcal{A}^{j_+} & \xrightarrow{\theta_{j_+}} & \mathcal{A} \\
& \searrow \gamma \times \text{id} & \downarrow \phi & \nearrow \theta_{j_+} & \\
& & \mathcal{Q}_G(i_+) \times \mathcal{A}^{i_+} & \xrightarrow{\theta_{i_+}} & \mathcal{A}
\end{array}$$

equals

$$\begin{array}{ccccc}
\mathcal{Q}_G(k) \times \prod_r (\mathcal{Q}_G(j_r) \times \prod_s (\mathcal{Q}_G(i_{r,s}) \times \mathcal{A}^{i_{r,s}})) & \xrightarrow{\text{id} \times \prod (\text{id} \times \theta_{i_{r,s}})} & \mathcal{Q}_G(k) \times \prod_r (\mathcal{Q}_G(j_r) \times \mathcal{A}^{j_r}) & \xrightarrow{\text{id} \times \prod \theta_{j_r}} & \mathcal{Q}_G(k) \times \mathcal{A}^k \\
\downarrow \gamma \times \text{id} & \searrow \text{id} \times \prod (\gamma \times \text{id}) & \downarrow \text{id} \times \prod \phi & \searrow \text{id} \times \prod \theta_{j_r} & \\
\mathcal{Q}_G(j_+) \times \prod_{r,s} (\mathcal{Q}_G(i_{r,s}) \times \mathcal{A}^{i_{r,s}}) & & \mathcal{Q}_G(k) \times \prod_r (\mathcal{Q}_G(i_r) \times \mathcal{A}^{i_r}) & \xrightarrow{\text{id} \times \prod \gamma} & \mathcal{Q}_G(k) \times \mathcal{A}^k \\
& \searrow \gamma \times \text{id} & \downarrow \gamma \times \text{id} & \nearrow \phi & \downarrow \theta_k \\
& & \mathcal{Q}_G(i_+) \times \mathcal{A}^{i_+} & \xrightarrow{\theta_{i_+}} & \mathcal{A}
\end{array}$$

(commuting squares are operadic associativity)

(iv) **(Operadic Identity)** The following two pairs of two pasting diagrams are equal:

$$\begin{array}{ccccc}
\mathcal{Q}_G(k) \times \mathcal{A}^k & \xrightarrow{\text{id} \times \iota^k} & \mathcal{Q}_G(k) \times (\mathcal{Q}_G(1) \times \mathcal{A})^k & \xrightarrow{\text{id} \times \theta_1^k} & \mathcal{Q}_G(k) \times \mathcal{A}^k \\
& & \downarrow \text{shuffle} & \nearrow \phi & \downarrow \theta_k \\
& & \mathcal{Q}_G(k) \times \mathcal{Q}_G(1)^k \times \mathcal{A}^k & & \\
& & \downarrow \gamma \times \text{id} & \nearrow \phi & \\
& & \mathcal{Q}_G(k) \times \mathcal{A}^k & \xrightarrow{\theta_k} & \mathcal{A}
\end{array}$$

equals

$$\begin{array}{ccccc}
& & \mathcal{Q}_G(k) \times (\mathcal{Q}_G(1) \times \mathcal{A})^k & & \\
& \nearrow \text{id} \times \iota^k & \downarrow \text{id} \times \Pi \phi & \searrow \text{id} \times \theta_1^k & \\
\mathcal{Q}_G(k) \times \mathcal{A}^k & \xrightarrow{\text{id}} & \mathcal{Q}_G(k) \times \mathcal{A}^k & \xrightarrow{\theta_k} & \mathcal{A} \\
& \downarrow \text{id} \times \iota^k & \nearrow \iota^k & \searrow \gamma \times \text{id} & \\
& \mathcal{Q}_G(k) \times (\mathcal{Q}_G(1) \times \mathcal{A})^k & \xrightarrow[\text{shuffle}]{} & \mathcal{Q}_G(k) \times \mathcal{Q}_G(1)^k \times \mathcal{A}^k &
\end{array}$$

(uses operadic associativity)

Also:

$$\begin{array}{ccccc}
\mathcal{Q}_G(k) \times \mathcal{A}^k & \xrightarrow{\iota} & \mathcal{Q}_G(1) \times \mathcal{Q}_G(k) \times \mathcal{A}^k & \xrightarrow{\text{id} \times \theta_k} & \mathcal{Q}_G(1) \times \mathcal{A} \\
& \searrow \text{id} & \downarrow \gamma \times \text{id} & \nearrow \phi & \downarrow \theta_1 \\
& & \mathcal{Q}_G(k) \times \mathcal{A}^k & \xrightarrow{\theta_k} & \mathcal{A}
\end{array}$$

equals

$$\begin{array}{ccc}
\mathcal{Q}_G(k) \times \mathcal{A}^k & \xrightarrow{\iota} & \mathcal{Q}_G(1) \times \mathcal{Q}_G(k) \times \mathcal{A}^k \\
\downarrow \theta_k & & \downarrow \text{id} \times \theta_k \\
\mathcal{A} & \xrightarrow{\iota} & \mathcal{Q}_G(1) \times \mathcal{A} \\
& \searrow \phi_1 & \downarrow \theta_1 \\
& & \mathcal{A}
\end{array}$$

(Uses operadic unitality)

Definition 6.7 ($\mathcal{Q}_G$ -pseudomorphism). A $\mathcal{Q}_G$ -pseudomorphism $(\mathbb{F}, \zeta) : (\mathcal{A}, \theta, \phi) \rightarrow (\mathcal{B}, \vartheta, \varphi)$ consists of functor $\mathbb{F} : \mathcal{A} \rightarrow \mathcal{B}$, such that for each j in the folloigin diagram commutes up to an invertible natural transformation ζ_j :

$$\begin{array}{ccc}
\mathcal{Q}_G(j) \times \mathcal{A}^j & \xrightarrow{\text{id} \times \mathbb{F}^j} & \mathcal{Q}_G(j) \times \mathcal{B}^j \\
\downarrow \theta_j & \nearrow \zeta_j & \downarrow \vartheta_j \\
\mathcal{A} & \xrightarrow{\mathbb{F}} & \mathcal{B}
\end{array}$$

(Guess: Want to not restrict ourselves to only equivariant 1-cells, if want so restrict to ones commuting with G -action)

Moreover, they satisfy the following properties:

- (i) **(Equivariance of ζ)** For each $j, \zeta_j = \zeta_j \circ ((g, \sigma) \times (g, \sigma)^{-1})$ for all $(g, \sigma) \in G \times \Sigma_j$.
- (ii) **(Preserves Identity)** The following two pasting diagrams are equal:

$$\begin{array}{ccc}
 \begin{array}{c}
 \mathcal{A} \xrightarrow{\mathbb{F}} \mathcal{B} \\
 \downarrow \iota \quad \quad \downarrow \iota \\
 \mathcal{Q}_G(1) \times \mathcal{A} \xrightarrow{\text{id} \times \mathbb{F}} \mathcal{Q}_G(1) \times \mathcal{B} \xrightarrow{\varphi_1} \mathcal{B} \\
 \searrow \theta_1 \quad \quad \uparrow \zeta_1^{-1} \quad \quad \searrow \vartheta_1 \\
 \mathcal{A} \xrightarrow{\mathbb{F}} \mathcal{B}
 \end{array}
 & = &
 \begin{array}{c}
 \mathcal{A} \xrightarrow{\mathbb{F}} \mathcal{B} \\
 \downarrow \iota \quad \quad \downarrow \text{id} \\
 \mathcal{Q}_G(1) \times \mathcal{A} \xrightarrow{\theta_1} \mathcal{A} \xrightarrow{\mathbb{F}} \mathcal{B}
 \end{array}
 \end{array}$$

- (iii) **(Preserves Composition)** The following two pasting diagrams are equal:

$$\begin{array}{ccccc}
 \mathcal{Q}_G(k) \times \prod_r (\mathcal{Q}_G(j_r) \times \mathcal{A}^{j_r}) & \xrightarrow{\text{id} \times \prod (\text{id} \times \mathbb{F}^{j_r})} & \mathcal{Q}_G(k) \times \prod_r (\mathcal{Q}_G(j_r) \times \mathcal{B}^{j_r}) & & \\
 \downarrow \gamma & \searrow \text{id} \times \prod \theta_{j_r} & \swarrow \text{id} \times \prod \zeta_{j_r} & \searrow \text{id} \times \prod \vartheta_{j_r} & \\
 & \mathcal{Q}_G(k) \times \mathcal{A}^k & \xrightarrow{\text{id} \times \mathbb{F}^k} & \mathcal{Q}_G(k) \times \mathcal{B}^k & \\
 \swarrow \phi & \downarrow \theta_k & \searrow \zeta_k & \downarrow \vartheta_k & \\
 \mathcal{Q}_G(j) \times \mathcal{A}^j & \xrightarrow{\theta_j} & \mathcal{A} & \xrightarrow{\mathbb{F}} & \mathcal{B}
 \end{array}$$

equals

$$\begin{array}{ccccc}
 \mathcal{Q}_G(k) \times \prod_r (\mathcal{Q}_G(j_r) \times \mathcal{A}^{j_r}) & \xrightarrow{\text{id} \times \prod (\text{id} \times \mathbb{F}^{j_r})} & \mathcal{Q}_G(k) \times \prod_r (\mathcal{Q}_G(j_r) \times \mathcal{B}^{j_r}) & & \\
 \downarrow \gamma & \searrow \text{id} \times \prod \theta_{j_r} & \swarrow \text{id} \times \prod \vartheta_{j_r} & \searrow \text{id} \times \prod \zeta_{j_r} & \\
 & \mathcal{Q}_G(j) \times \mathcal{A}^j & \xrightarrow{\text{id} \times \mathbb{F}^j} & \mathcal{Q}_G(j) \times \mathcal{B}^j & \\
 \swarrow \theta_j & \downarrow \zeta_j & \searrow \vartheta_j & \downarrow \vartheta_k & \\
 \mathcal{A} & \xrightarrow{\mathbb{F}} & \mathcal{B} & &
 \end{array}$$

Definition 6.8 ($\mathcal{Q}_G$ -2-cells). Let $(\mathbb{F}, \zeta), (\mathbb{G}, \xi) : (\mathcal{A}, \theta, \phi) \rightarrow (\mathcal{B}, \vartheta, \varphi)$ be two $\mathcal{Q}_G$ -pseudomorphisms, a $\mathcal{Q}_G$ -2-cell $\lambda : (\mathbb{F}, \zeta) \Rightarrow (\mathbb{G}, \xi)$ is a natural transformation $\lambda : \mathbb{F} \rightarrow \mathbb{G}$, such that the following two pasting diagrams

are equal:

$$\begin{array}{ccc}
 & \text{id} \times \mathbb{F}^j & \\
 \mathcal{Q}_G(j) \times \mathcal{A}^j & \xrightarrow{\text{id} \times \lambda^j} & \mathcal{Q}_G(j) \times \mathcal{B}^j \\
 \theta_j \downarrow & \text{id} \times \mathbb{G}^j \nearrow & \downarrow \vartheta_j \\
 \mathcal{A} & \xrightarrow{\xi_j} & \mathcal{B} \\
 & \mathbb{G} &
 \end{array}
 =
 \begin{array}{ccc}
 & \text{id} \times \mathbb{F}^j & \\
 \mathcal{Q}_G(j) \times \mathcal{A}^j & \xrightarrow{\text{id} \times \lambda^j} & \mathcal{Q}_G(j) \times \mathcal{B}^j \\
 \theta_j \downarrow & \zeta_j \nearrow & \downarrow \vartheta_j \\
 \mathcal{A} & \xrightarrow{\lambda} & \mathcal{B} \\
 & \mathbb{G} &
 \end{array}$$

6.3 Category $\mathcal{F}_G$ (ND)

Definition 6.9. $\mathcal{F}_G$ denotes the *category of based finite sets*, where the objects are $\mathbf{n}^\alpha$, set wise being the finite set $\{0, 1, \dots, n\}$ with 0 fixed as the basepoint, and $\alpha : G \rightarrow \Sigma_n$ is a based G -action on the set.

The homset $\mathcal{F}_G(\mathbf{n}^\alpha, \mathbf{m}^\beta)$ consists of all the based set map, which the homset also endows with a G -action via conjugation. From now on, we'll view $\mathcal{F}_G$ as a 2-category, where each hom-category $\mathcal{F}_G(\mathbf{n}^\alpha, \mathbf{m}^\beta)$ is discrete with only identity 2-cells.

Furthermore, we define the following:

- $\Pi_G \subset \mathcal{F}_G$ denotes the subcategory where each morphism $\phi : \mathbf{n}^\alpha \rightarrow \mathbf{m}^\beta$ satisfies $|\phi^{-1}(j)| = 0, 1$ for all $j \neq 0$.
- $\Lambda_G, \Lambda_G^\perp, \Sigma_G \subset \Pi_G$ denotes the subcategory where each morphism is injective, surjective, and bijective respectively (in particular, $\Lambda_G \cap \Lambda_G^\perp = \Sigma_G$).
- $\mathcal{E}_G \subset \mathcal{F}_G$ denotes the subcategory where each morphism satisfies $\phi^{-1}(0) = \{0\}$.

6.4 2-Category $\mathcal{F}_G\text{-PsAlg}$

Definition 6.10 ($\mathcal{F}_G$ -pseudoalgebra). An $\mathcal{F}_G$ -pseudoalgebra, is a pseudofunctor $\tilde{\mathcal{A}} : \mathcal{F}_G \rightarrow \mathbf{Cat}_G$, such that restricting to the subcategory $\Pi_G \subset \mathcal{F}_G$, $\tilde{\mathcal{A}}$ is a strict functor. Moreover, we require it to be reduced, namely $\tilde{\mathcal{A}}(0) = *$ (the terminal G -category), and each hom-functor

$$\tilde{\mathcal{A}}_{\mathbf{n}^\alpha, \mathbf{m}^\beta} : \mathcal{F}_G(\mathbf{n}^\alpha, \mathbf{m}^\beta) \rightarrow \mathbf{Cat}_G(\tilde{\mathcal{A}}(\mathbf{n}^\alpha), \tilde{\mathcal{A}}(\mathbf{m}^\beta))$$

is a G -functor.

Question: In operad case, we have natural isomorphism between any two G -actions, do we need it here?

Let $\delta_i : \mathbf{n}^\alpha \rightarrow \mathbf{1}$ be the i th projection (namely $i \mapsto 1$, and any $i \neq j \mapsto 0$), since it belongs to Π_G , it generates a functor $\tilde{\mathcal{A}}\delta_i : \tilde{\mathcal{A}}(\mathbf{n}^\alpha) \rightarrow \tilde{\mathcal{A}}(\mathbf{1})$. Hence, by universality of product, all $\tilde{\mathcal{A}}\delta_i$ ($1 \leq i \leq n$) induces a functor $\delta : \tilde{\mathcal{A}}(\mathbf{n}^\alpha) \rightarrow \tilde{\mathcal{A}}(\mathbf{1})^n$.

Definition 6.11 ($\mathcal{F}_{G,\text{vs}}$ and $\mathcal{F}_{G,\mathbb{R}}$ -pseudoalgebra). An $\mathcal{F}_G$ -pseudoalgebra $\tilde{\mathcal{A}}$ is *very special* (or an $\mathcal{F}_{G,\text{vs}}$ -pseudoalgebra), if all $\delta : \tilde{\mathcal{A}}(\mathbf{n}^\alpha) \rightarrow \tilde{\mathcal{A}}(\mathbf{1})^{\mathbf{n}^\alpha}$ is an isomorphism for all $\mathbf{n}^\alpha \in \mathcal{F}_G$, and $\tilde{\mathcal{A}}$ is *strictly special* (or an $\mathcal{F}_{G,\mathbb{R}}$ -pseudoalgebra), if all δ is in fact identity.

Definition 6.12 ($\mathcal{F}_G$ -1-cell). Given $\tilde{A}, \tilde{B}$ two $\mathcal{F}_G$ -pseudoalgebras, a morphism (or $\mathcal{F}_G$ -1-cell) between them is a pseudo-transformation $(\mathbb{F}, \zeta) : \tilde{\mathcal{A}} \rightarrow \tilde{\mathcal{B}}$, such that restricting to Π_G , it becomes an actual natural transformation.

Definition 6.13 ($\mathcal{F}_G$ -2-cell). Given $\tilde{A}, \tilde{B}$ two $\mathcal{F}_G$ -pseudoalgebras, and two $\mathcal{F}_G$ -1-cells $(\mathbb{F}, \zeta), (\mathbb{G}, \xi) : \tilde{A} \rightarrow \tilde{B}$, an $\mathcal{F}_G$ -2-cell is a transformation $\lambda : \mathbb{F} \Rightarrow \mathbb{G}$, such that for any morphism $f : \mathbf{n}^\alpha \rightarrow \mathbf{m}^\beta$ in $\mathcal{F}_G$, the following two pasting diagrams are equal:

$$\begin{array}{ccc}
 \begin{array}{ccc}
 & \xrightarrow{\mathbb{F}_{\mathbf{n}^\alpha}} & \\
 \tilde{\mathcal{A}}(\mathbf{n}^\alpha) & \xRightarrow{\lambda} & \tilde{\mathcal{B}}(\mathbf{n}^\alpha) \\
 \tilde{\mathcal{A}}f \downarrow & \swarrow \mathbb{G}_{\mathbf{n}^\alpha} \quad \searrow \xi f & \downarrow \tilde{\mathcal{B}}f \\
 \tilde{A}(\mathbf{m}^\beta) & \xrightarrow{\mathbb{G}_{\mathbf{m}^\beta}} & \tilde{B}(\mathbf{m}^\beta)
 \end{array}
 & = &
 \begin{array}{ccc}
 & \xrightarrow{\mathbb{F}_{\mathbf{n}^\alpha}} & \\
 \tilde{\mathcal{A}}(\mathbf{n}^\alpha) & \xrightarrow{\quad} & \tilde{\mathcal{B}}(\mathbf{n}^\alpha) \\
 \tilde{\mathcal{A}}f \downarrow & \swarrow \zeta f \quad \searrow \mathbb{F}_{\mathbf{m}^\beta} & \downarrow \tilde{\mathcal{B}}f \\
 \tilde{A}(\mathbf{m}^\beta) & \xRightarrow{\lambda} & \tilde{B}(\mathbf{m}^\beta) \\
 & \searrow \mathbb{G}_{\mathbf{m}^\beta} &
 \end{array}
 \end{array}$$

7 Construction of $\mathbb{R}_G$ (ND)

7.1 Construction of Objects

7.2 Extension to 2-functor

Reference Not Done (find their corresponding journal, if not the arxiv)

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